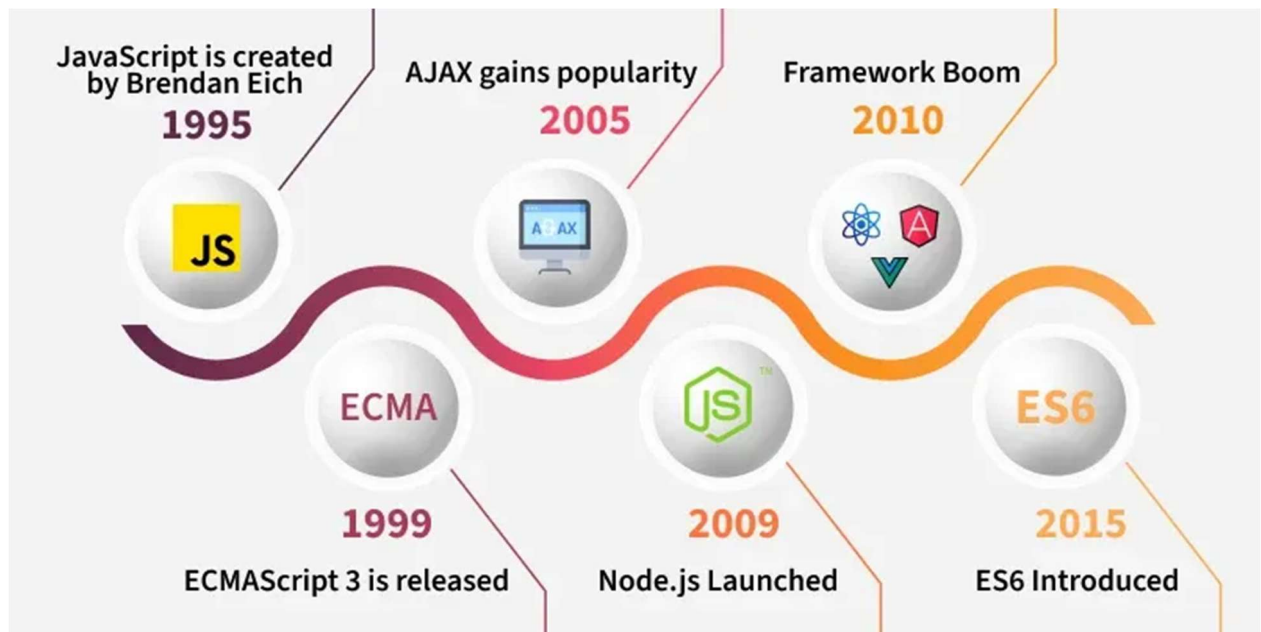




## What is Javascript ?

JavaScript is a high-level, interpreted programming language that is one of the core technologies of the World Wide Web, alongside HTML and CSS. While HTML provides the structure of web pages and CSS specifies their style, JavaScript adds interactivity and dynamic behavior, transforming static pages into dynamic applications.

### How to Add JavaScript in HTML Document?

To add JavaScript in HTML document, several methods can be used. These methods include embedding JavaScript directly within the HTML file or linking an external JavaScript file.

#### Inline JavaScript

You can write JavaScript code directly inside the HTML element using the onclick, onmouseover, or other event handler attributes.

#### Internal JavaScript (Within <script> Tag)

You can write JavaScript code inside the <script> tag within the HTML file. This is known as internal JavaScript and is commonly placed inside the <head> or <body> section of the HTML document.

## External JavaScript (Using External File)

For larger projects or when reusing scripts across multiple HTML files, you can place your JavaScript code in an external .js file. This file is then linked to your HTML document using the src attribute within a <script> tag.

## JavaScript Values

The JavaScript syntax defines two types of values:

- **Fixed values:** The fixed values are known as literals.
- **Variable values:** The variable values are called variables.

## Literal In JS

In JavaScript, a literal is a fixed value that you write directly into your source code. It's not a variable; it's the actual data itself.

For example, in the statement `let x = 10;`, 10 is a numeric literal.

### **Common Literal Types**

JavaScript has literals for its primitive data types and for objects.

### String Literals

These represent a sequence of characters and are enclosed in quotes.

- Single quotes: 'Hello, world!'
- Double quotes: "This is a string."
- Template literals (backticks): `You can embed expressions like  $\{2 + 2\}$  in these.`

### Numeric Literals

These represent numbers. They can be integers, floating-point numbers, or in scientific notation.

```
let age = 25;      // 25 is an integer literal
```

```
let price = 19.99; // 19.99 is a floating-point literal
```

```
let largeNumber = 5e6; // 5e6 (5 million) is an exponential literal
```

## **Boolean Literals**

These represent a logical entity and can only have one of two values.

```
let isLoggedIn = true; // true is a boolean literal
```

```
let isFinished = false; // false is a boolean literal
```

## **Array Literals**

An array literal is a list of zero or more expressions, enclosed in square brackets [].

```
let emptyArray = []; // An empty array literal
```

```
let fruits = ["Apple", "Banana", "Cherry"]; // An array literal with strings
```

```
let numbers = [1, 2, 3, 4, 5]; // An array literal with numbers
```

## **Object Literals**

An object literal is a list of zero or more key-value pairs, enclosed in curly braces {}. It's a very common way to create objects in JavaScript.

```
let emptyObject = {}; // An object literal
```

## **JavaScript console.log()**

In JavaScript, the console.log() method displays messages or variables in the browser's console.

### **Syntax**

```
console.log(message);
```

### **Example**

```
let data = "Hello, JavaScript!";
```

```
console.log(data);
```

```
// Output: Hello, JavaScript!
```